

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.2: Area of Polygons — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.2: Area of Polygons
Driving Question	DRIVING QUESTION: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Area of a Triangle ($\frac{1}{2} \times \text{base} \times \text{height}$)	Learners observed a sketch of an irregular five-sided shamba in Nakuru County with five unequal side lengths and no perpendicular heights marked. They observed three triangles on squared paper — one with an easy right-angle height, one with a drawn perpendicular height, and one with NO height drawn — and used counting and the formula $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ to find each triangle's area.	Learners (re)established that the formula for the area of a triangle is $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$, and that the height MUST be the perpendicular distance from the base to the opposite vertex. They learned that when a triangle is drawn in a non-right-angle orientation, the perpendicular height must be constructed — it is not the same as any side length. They connected this requirement directly to the surveyor's problem: the shamba cannot be measured with this formula alone because no perpendicular heights are accessible.	Learners explained (in groups and in their individual Growing Model) that Surveyor Wanjiku cannot directly apply the formula $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ to the triangles in the shamba because the perpendicular heights are not measurable on uneven ground with maize crops. They articulated this as a precise knowledge gap: to calculate the area of an irregular polygon by decomposing it into triangles, a method for finding triangle area WITHOUT a perpendicular height is needed. This gap launches the unit's investigation into Heron's formula and the sine area rule.
Lesson 2 Area of a Triangle Using the Sine Formula ($\frac{1}{2}ab \sin \theta$)	Learners observed a scale diagram of a triangular corner of the Nakuru shamba ($AB = 120 \text{ m}$, $AC = 95 \text{ m}$, angle $A = 68^\circ$) where a seasonal stream blocked direct measurement of the perpendicular height. They followed a step-by-step derivation showing how the perpendicular height h can be expressed as $h = c \sin A$ using trigonometry, transforming the standard formula into $A = \frac{1}{2}ab \sin \theta$. They also observed that in Problem 2, a small	Learners learned that when the perpendicular height of a triangle is not directly measurable but two sides and the included angle are known, the area can be calculated exactly using $A = \frac{1}{2}ab \sin \theta$. They learned that the angle in this formula must be the included angle — the angle formed between the two known sides — and that the size of this angle critically determines the area. They also learned that this formula is derived directly from the	Learners explained that the Nakuru surveyor can now calculate the area of any triangular section of the five-sided shamba where two side lengths and the included angle are recorded in her field notes, even if she could not physically measure the height due to the stream. They explained why the assistant in Problem 2 was wrong — longer sides do not guarantee a larger area if the included angle is small. They updated their cumulative shamba model by shading and

	included angle (30°) produced a surprisingly small area — smaller than a nearby rectangle — even with sides of 60 m and 90 m.	standard $\frac{1}{2} \times \text{base} \times \text{height}$ formula using the trigonometric identity $h = b \sin \theta$, meaning no new mathematics is invented — only existing tools are combined in a new way.	labelling Triangle ABC (area $\approx 5\,285\text{ m}^2$) and articulated that the sine formula is the first of multiple tools needed to decompose and solve the entire irregular polygon.
Lesson 3 Applying the Sine Formula for Area of a Triangle	Learners watched a narrated video showing a builder in Eldoret and a fisherman on Lake Victoria each calculate a triangle's area using the sine formula when no perpendicular height was available. They recorded structured observation notes for both examples.	Learners learned that when two sides and the included angle of a triangle are known, the area can be calculated using $A = \frac{1}{2}ab \sin C$, because the perpendicular height can be expressed as $h = a \sin C$. They practised selecting between this formula and $\frac{1}{2}bh$ based on what information is given.	Learners explained their formula selections in writing and connected the sine formula to the Nakuru surveyor's situation — she can now calculate the area of any triangle within the shamba where she has measured two boundary sides and the angle between them, bringing her one step closer to finding the total area of the five-sided plot.
Lesson 4 Area of a Triangle Using Heron's Formula	Learners observed a step-by-step derivation showing how Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$ emerges algebraically from the cosine rule combined with the sine area formula. They also observed that the Nakuru surveyor's northern triangular sub-plot (sides 40 m, 55 m, 63 m) and the Kisumu carpenter's triangular panel (sides 7 cm, 9 cm, 11 cm) both lack height and angle data — making previously learned formulas insufficient without Heron's formula.	Learners learned that: (1) the semi-perimeter $s = (a+b+c)/2$ is the essential first step in Heron's formula; (2) Heron's formula calculates the exact area of any triangle given only its three side lengths; (3) Heron's formula and the sine formula are consistent — they yield the same area when both are applicable, which allows cross-checking; (4) real-world applications (e.g., calculating fertiliser bags for a Kericho maize farm) require rounding area-based answers UP to ensure sufficiency.	Learners explained that the Nakuru land surveyor can now calculate the area of any triangular sub-plot of the shamba even when only fence-post distances (side lengths) are recorded and no angles or heights are available, by applying Heron's formula. They updated their cumulative shamba decomposition model to label the northern triangular sub-plot as a 'Heron's formula triangle,' and articulated in writing the specific condition that makes Heron's formula the appropriate tool: three side lengths known, no height or angle available.
Lesson 5 Polygons: Classification, Properties, and Decomposition into Triangles	Learners sorted 10 pre-cut polygon shapes (regular/irregular, convex/concave) onto a four-quadrant sorting mat and physically cut a paper irregular pentagon into triangles by drawing diagonals from a single vertex, counting 3 triangles. They repeated this with a hexagon, finding 4 triangles.	Any polygon with n sides can be decomposed into $(n - 2)$ non-overlapping triangles by drawing diagonals from one vertex. For the five-sided shamba plot in Nakuru, this gives 3 triangles. The total area of the polygon equals the sum of the areas of these triangles, each calculated using either the sine area formula or Heron's formula depending on the available measurements.	The surveyor cannot use a single formula directly on the five-sided irregular shamba plot because it is not a standard shape. By classifying it as an irregular convex pentagon and decomposing it into 3 triangles (ABC, ACD, ADE), the surveyor can apply the triangle area tools from Lessons 1–4 to each triangle separately and add the results. This decomposition strategy works for ANY polygon, making it a universal tool for land area problems.
Lesson 6 Area of Regular Polygons: Central Triangle Decomposition	Learners constructed regular hexagons and pentagons, drew radii from the centre to each vertex, and observed that every regular polygon decomposes into n congruent isosceles triangles. They measured the apothem and central angle of each triangle, and recorded data in a	Learners derived the general formula for the area of any regular polygon: $A = (n \times s^2) / (4 \times \tan(180^\circ/n))$, where n is the number of sides and s is the side length. They applied the formula to calculate the area of hexagonal floor tiles at Nairobi's Jamia Mosque ($s = 20\text{ cm}$), a pentagonal flower	Learners explained that the regular polygon formula works because perfect symmetry allows every polygon to be split into identical central triangles, each with a calculable area using only the side length and the tangent of the half-central-angle. However, they also explained that this

	structured table — noting that the central angle always equals $360^\circ/n$ and that the apothem relates to the side length via $\text{apothem} = (s/2) / \tan(180^\circ/n)$.	bed at the Nairobi Arboretum ($s = 3.5$ m), and a regular octagonal boda-boda waiting bay in Kisumu ($s = 2$ m).	formula CANNOT be directly applied to the Nakuru surveyor's irregular shamba, because the shamba's sides are all unequal and its angles are not congruent — confirming the need for the irregular polygon decomposition strategy to be developed in Lesson 7.
Lesson 7 Area of Irregular Polygons: Decomposing the Surveyor's Shamba	We observed that the surveyor's five-sided shamba in Nakuru — and both the agricultural officer's quadrilateral plot near Eldoret and the pentagonal mural wall at the Nakuru school — could not be solved as a single shape using one formula. When we drew diagonals from one vertex, the irregular polygons broke apart into familiar triangles. We observed that different triangles within the same polygon sometimes required different formulae depending on which measurements were available.	We learned that any irregular polygon can be decomposed into non-overlapping triangles by drawing diagonals from a single vertex. The number of triangles formed is always two fewer than the number of sides ($n - 2$). We learned to select the correct area formula for each triangle based on available data: $\frac{1}{2}ab \sin C$ when two sides and the included angle are known, Heron's formula ($A = \sqrt{s(s-a)(s-b)(s-c)}$) when all three sides are known, and $\frac{1}{2} \times \text{base} \times \text{height}$ when a perpendicular height is available. The total area of the irregular polygon equals the sum of all triangle areas.	We explained that the land surveyor in Nakuru can calculate the exact area of the five-sided shamba by drawing two diagonals (AC and AD) from vertex A, decomposing the pentagon into triangles ABC, ACD, and ADE. For each triangle, the surveyor applies whichever formula matches the available field measurements. Summing the three triangle areas gives the total plot area in square metres. This decomposition strategy works for any irregular polygon, regardless of the number of sides, making it the universal tool in the surveyor's mathematical toolkit.
Lesson 8 Consolidation, Model Revision & Real-World Applications: Answering the Driving Question	Students observed (across the unit) that the Nakuru surveyor could not apply the standard $\frac{1}{2} \times \text{base} \times \text{height}$ formula to the irregularly shaped five-sided shamba because no perpendicular height was directly measurable from the given field data. They also observed that the plot could be physically decomposed into triangles and quadrilaterals by drawing internal diagonals, making the problem tractable.	Students learned that any polygon area problem can be solved by selecting from a toolkit of five strategies: (1) $\frac{1}{2}bh$ when a perpendicular height is available; (2) $A = \frac{1}{2}ab \sin C$ when two sides and an included angle are known; (3) Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$ when all three sides of a triangle are known but no angle or height is given; (4) specific quadrilateral formulas for parallelograms, trapezoids, and kites; and (5) $A = ns^2/[4 \tan(180^\circ/n)]$ for regular polygons. Irregular polygons are solved by decomposing them into these known shapes and summing the component areas.	Students can now explain that a land surveyor in Nakuru can calculate the exact area of any irregularly shaped farm plot by (a) identifying which measurements are available at each boundary triangle or quadrilateral after decomposing the plot with internal diagonals, (b) selecting the appropriate formula for each component shape based on those available measurements, (c) calculating each component area and summing them, and (d) converting units to hectares for practical land use. The surveyor does not need a perpendicular height for every triangle — the sine formula and Heron's formula eliminate that requirement entirely, making the complete measurement of any polygon possible with the toolkit developed in this unit.

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